

Ensuring US Leadership and Security in AI through Neuromorphic and Photonic Implementations of Artificial Intelligence

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Introduction

The Office of Science and Technology Policy (OSTP) has issued a *Request for Information on the Development of an Artificial Intelligence (AI) Action Plan*. We anticipate this RFI will receive numerous submissions that explore strategies to advance mainstream artificial intelligence and machine learning approaches that promise to grow a national workforce trained in AI, and that propose the deployment of a wide array of infrastructure investments to maintain the US's leadership in AI. The purpose of our submission, which represents the view of three private citizens, is to present two important problems that might not be adequately represented in other submissions, as well as two separate technologies that the authors of this response believe deserve a place in the nation's AI toolbox. Lastly, we pull these two technologies together into a final opportunity for the AI Action Plan that aims to combine both of their advantages.

The two concerns that we believe need to be addressed are:

- The extremely large amount of computation and therefore energy needed for the most powerful forms of AI,
- The national security risks associated with AI implementations reliant on electronic components, such as semiconductor-based computing.

To address these two unique challenges, we believe that there are two distinct technologies that the nation should pursue in parallel with advancing AI's current technological trends. Specifically, we believe that neuromorphic computing, and implementations of AI primitives and fully operational AI algorithms in photonic circuitry are cutting-edge technologies that the US must support in the future.

This document will provide a brief overview of the two challenges and then identify the opportunities that can arise from investing in neuromorphic computing and photonic AI/ML implementations. For both approaches, we will present recommendations on how the nation can advance the development of these complementary technologies.

Computing, Energy and National Security Risks

The abundance of data being collected and made available to the wide array of algorithms that we will rely upon in our daily living will increase rapidly, especially as we continue to deploy more sensors and actuators in our world in an effort to improve how our nation operates. Unfortunately, the sheer volume of data and increasingly prohibitive amount of computing needed for training extremely large-scale AI/ML algorithms will eventually place significant stress upon both the nation's computational infrastructure, as well as the nation's power grid. The US has witnessed several noteworthy failures in its power grid infrastructure. The 2003 blackout, which hit the Northeast and Midwest regions, was attributed to software malfunctions and high demand, placing catastrophic stress on the nation's aging electricity grid. Another set of examples were the rolling blackouts in California in 2006 and 2020, which resulted from surges in demand driven by high summer temperatures. Taken together, the power grid failures that the nation has experienced indicate that *the US needs to make every effort to reduce the stress*

that a projected surge in AI computations could place upon our power grid. Meanwhile, the nation must also invest in a next-generation power grid that can reliably meet the surging electricity demand driven by growing consumer usage and the rapid deployment of nation-scale data and computational centers.

While a nationwide increase in computing could place significant stress upon our nation's power delivery infrastructure, and lead to increases in utility prices for the average American citizen, there is a separate risk that the nation faces if we become entirely reliant on computing based on conventional semiconductor electronics. Electronic components emit electromagnetic emanations that can be subject to various forms of eavesdropping and electronic espionage directed against our nation's computing and telecommunications infrastructure. As more of our nation's important decisions and analyses utilize AI algorithms, the corresponding communication and computational infrastructure must be protected from nation-state adversaries that monitor the US's innovation and business operations. Further, electronics-based computing is also fragile and susceptible to intentional use of electromagnetics that can disrupt and damage our computing investments. An adversarial application of high-power electromagnetic emissions (EMPs) has the potential to induce currents in circuitry that exceed the tolerance levels of electronic components. This could result in the failure of our nation's investments in high-performance computational infrastructure for artificial intelligence.

Opportunity-1: Invest in Neuromorphic AI

The predominant approaches to AI algorithms, such as large language models and convolutional neural networks, require enormous computing power. Eventually, in a future where the computing requirements for conventional mainstream AI continue to grow, there will be many applications that cannot utilize computationally expensive, data-rich approaches to AI. This will leave the US behind in developing the full spectrum of AI solutions that our nation requires. Therefore, there is an urgent need to develop alternative versions of AI/ML that achieve significant improvements in algorithmic efficiency, dramatically reduce energy requirements, and which could maintain US advancement across the full spectrum of data-driven AI applications.

One philosophical view for how advancements in AI development has progressed in the past decade is that researchers have been focused on developing bigger and better algorithms while throwing larger and faster computation resources at the problem. Although this has led to many noteworthy successes, it also fails to leverage the evolutionary insights that nature provides in terms of how we could design better and naturally efficient approaches to implementing artificial intelligence and machine learning algorithms.

The human brain is a highly advanced "biological computer" that is also energy-efficient: it is able to perform complex calculations while consuming less power than a conventional light bulb. It achieves this through an optimized combination of sparse activation, parallel processing, adaptability, and analog-style computation. These advantages have led to an alternate approach to computing, known as *neuromorphic* systems, which use hardware and algorithms that emulate human neurons and their known advantages. Neuromorphic computation could lead to a significant advancement in AI by making it more energy-efficient and suitable for solving problems that require real-time intelligence. In short, neuromorphic computing offers a huge advantage for the US to invest in.

The current generation of neuromorphic chips, as illustrated by Intel's Loihi and IBM's TrueNorth, are based on spiking neural networks, which only activate the neurons when needed. This fact implies that unnecessary calculations are not performed, and would mean that AI based on neuromorphic computing can be orders of magnitude more scalable. One approach to neuromorphic computing uses spike-timing-dependent plasticity (STDP), a biological learning rule, which is a promising candidate for next-generation AI models because of its energy-efficiency and its ability to support real-time learning in future AI implementations.

Recommendations for a National AI Action Plan:

- *Launch Milestone-oriented National Challenges:* To support rapid development and breakthroughs in neuromorphic AI, we believe the nation should build upon the model of DARPA Grand Challenges and hold contests with specific high-stretch performance milestones (e.g. energy efficiency or reduction in computation for a benchmark task). The prizes should incentivize taking bold risks for innovation.
- *Launch Prototype-oriented National Challenges:* A similar approach to the milestone challenges, but the focus here would be on giving a set of prototype goals, such as a lifelike robot that uses neuromorphic computing to learn from its sensor data and demonstrates improved performance over time.
- *Prepare Adjacent Industries for Rapid Integration:* There are many application areas that could benefit from neuromorphic AI advances, and the associated industries must be prepared to quickly integrate. Having strategies ready that encourage adoption, such as:
 - o Creating tax-incentives for adopting new neuromorphic technologies;
 - o Reducing licensing charges for initial product market launch;
 - o Hosting industry-day events that facilitate the connection between vendors with problems and vendors with solutionswould kick start industrial growth.
- *Defense Integration Plan:* There are many applications in the realm of national defense that could benefit from neuromorphic computing, such as advancing human-computer interfaces, robotics and human-assist augmentation. The nation should develop a separate action plan for how different AI technologies can be integrated into DoD and NASA applications.
- *Public-Private Think Tanks for AI:* To ensure that the nation has the best minds advancing neuromorphic computing approaches to AI, we would recommend the nation developing a "think tank" that is focused on supporting a broad array of approaches to AI, such as neuromorphic computing, privacy-preserving AI, and AI on quantum computing. The US needs to be ready to maintain research across the full spectrum of AI approaches to ensure it has a full repertoire for national competitiveness.

Opportunity-2: Invest in Photonic Implementations of AI

There are fundamental hurdles impeding the continued success of electronic-based computing for large-scale AI. Electronic circuitry built upon semiconductor technology is energy inefficient due to the inherent resistance-induced heating, architectural bottlenecks, and current leakage in the chips. As a result, AI implementations reliant on electronics will always carry the baggage of energy inefficiency and heat production. Other issues, such as the increase in power

densities for nanometer scale electronics and quantum tunneling point to limits in what can be accomplished through electronics, and will require expensive and energy-intensive approaches to cooling AI computational centers. We believe that the nation needs to strongly invest in alternative computing paradigms that can minimize the long-horizon burden upon our nation's energy infrastructure.

Computing based upon optical signals, or photonic computing, employs light signals as the basis for computation and offers a more natural approach to reducing energy than is possible with electronics. When transmitted through appropriate photonic materials, photons experience almost no resistance and thereby produce significantly less heat and energy loss than electronics. This fact also implies that photonic computing, when fully developed, can operate faster than electronics since photons have only weak interactions with their media, while electronic signals inherently experience interference in semiconductor materials. Further, light-based computation can utilize multiple wavelengths to achieve natural levels of parallelism, which contrasts with the serial nature of digital electronics. To summarize, strictly in terms of architectural advantages for computing, photonic computing has the following advantages over conventional electronics:

- Generates minimal heat,
- Easier thermal maintenance,
- Higher data rates,
- Easily parallelized.

Another significant advantage is that photonic implementations for computing would not suffer from the national security risks associated with electromagnetic phenomena:

- Photonic circuits generate almost no electromagnetic emanations, which makes them ideal for secure communications since they are essentially unsusceptible to Van Eck phreaking and other forms of electromagnetic eavesdropping.
- Photonic circuits are built using dielectric materials, which makes them inherently less susceptible to inducing large current spikes that are associated with electromagnetic pulses.

Consequently, photonic circuits offer increased resilience to eavesdropping and denial of service threats, making them an ideal platform for building new and secure AI algorithms.

Recommendations for a National AI Action Plan:

- *Support Research and Development in Miniaturization:* Photonic circuitry will benefit from advancements in miniaturization, and ultimately lead to reduced power requirements, lower manufacturing costs, and easier integration into applications.
- *Address National Photonic Supply Chain Constraints:* Photonic materials depend on rare earth materials and have high-purity requirements, which has led to challenges in the US supply chain for photonics. The US must ensure it has a complete photonics supply chain that is robust to geopolitical constraints.
- *Invest in Creating a Photonic Ecosystem:* Unlike the semiconductor industry, the photonic industry has yet to achieve the benefit of economies of scale. Investments in regional photonic innovation hubs, as well as in a manufacturing sector in support of photonics, would allow photonic AI implementations to move forward more easily.
- *Establish Software Industry for Photonic AI:* There is an urgent need to develop a software industry that can support the implementation of AI algorithms using photonic

devices, much like the software industry supports AI implementation on semiconductor computers.

- *Host National Competition for Data Loading and Access:* One challenge facing AI implementations, and particularly on photonic circuitry, is the disparity between processing speed and importing data from memory registers. The government should invest in innovative photonic solutions like architectures that co-locate memory with photonics, and optical memory technologies such as holographic data storage.

Opportunity-3: Photonic Neuromorphic AI

Neuromorphic approaches to implementing AI provide a technological path to achieving reductions in computational requirements, making it more energy-efficient and suitable for solving problems that require real-time intelligence. Photonic computing also provides a path to achieving reduction in heat and energy loss, provides unique security benefits, supports parallelized algorithm implementation, and ultimately faster computations. Pulling both technologies together into a single, unified approach to AI offers unique advantages that should be considered as part of the national AI Action Plan.

Neuromorphic photonic computing (NPC) explores both ultra-efficient brain-inspired photonic neural networks and ultrafast photonic hardware for computationally demanding machine learning (ML) techniques in modern artificial intelligence (AI). In the past decade, NPC has made a successful transition from free-space optics and fiber optics and discrete optical components to employing silicon-photonics-based integrated hardware. The community developing NPC solutions is primed and ready to embrace further technology breakthroughs and innovations in silicon photonic foundry platforms, photonic neural network microarchitectures, and more. It is expected that the further research and technology development of NPC in the next 5-10 years will result in transformative impacts on computing and solve many challenging problems in important applications in robotics, edge computing, physical system simulations, computer vision and avionics.

With silicon photonics, NPC leverages some of the ideal features of photonics, such as parallelism with multiple wavelengths, ultra-low latency with time-of-flight processing, high bandwidth with lightwave, and nonlinear dynamics to support scalable integration of complex photonic neural networks. The fundamental computing node of an NPC network is the photonic neuron. Each photonic neuron's key functions include independent weighting of multiple inputs, summation, and nonlinear transformation. These operations can be mapped and implemented on a silicon photonic integrated circuit using microring resonators (MRRs). Wavelength division multiplexing (WDM) enables lightwave signals at different wavelengths to share the same transmission medium (such as optical fiber or waveguide), and is a key commercial technology used in tele-communication networks. In a WDM-compatible photonic neural network, each of the input signals (data) are mapped to different wavelengths for wavelength-dependent weighting and parallel detection of multiple wavelengths for summation.

While developing photonic neural networks is an initial component, integrating these optical neurons together into a fully working system requires that additional technological hurdles are overcome. Some of these challenges include:

- Neuromorphic AI algorithms implement real-valued calculations, which introduce complexity in implementation, as well as when storing such calculations in optical memory

architectures.

- Algorithmic feedback, which is a foundational primitive to achieving adaptation in AI implementations, requires timing and synchronization to avoid instability in photonic implementations, thus requiring more careful engineering and design.
- Networks of photonic neurons introduce quadratic scalability challenges, leading to challenges in preventing crosstalk in physical layouts, as well as challenges in wavelength assignments for large neural networks.

As a result, neuromorphic photonic circuits introduce additional technical challenges when compared to either neuromorphic computing or photonic computing. These challenges, though, deserve to be pursued and overcome as part of the national AI Action Plan because the advantages of AI implemented through neuromorphic photonic circuits would provide the US a unique competitive edge in AI technologies.

Recommendations for a National AI Action Plan:

- *Establish Long-Horizon Federal Funding:* Neuromorphic photonic computing and implementing AI on NPC represents a longer-term technology investment, and likely will require a decade-long investment by a funding agency, such as DARPA, to fully address technology hurdles.
- *Support US Graduate Students in NPC:* The US must invest in developing an advanced and talented workforce that is trained in both photonics, computing, and artificial intelligence disciplines. To achieve this, the nation needs to expand the portfolio of Ph.D. fellowship programs that exist in AI, photonics, and neurocomputing.
- *Host National NPC Architecture Competition:* There is a specific need to develop the best photonic neural network microarchitectures for the NPC field to utilize in making further advancements. To foster such innovation, the nation should hold a national competition specifically focused on photonic neural network microarchitectures, with suitable prizes such as startup investments.
- *Improve SBIR/STTR Programs:* Currently, US SBIR/STTR funding programs have a flaw in how they are designed. Many innovative technologies and successful companies fail because of funding gaps that exist between program phases, causing the US to effectively lose its investments. The design of the SBIR/STTR funding ecosystem should be improved to foster the success and survivability of companies that have achieved good performance for their technology and hit target benchmarks.
- *Encourage Venture Capital Investment in NPC:* Venture capital investments are a powerful tool that the US can use to spur innovation in high-risk and high-payoff technologies like NPC and AI. The nation should encourage venture capital investments through mechanisms by providing matching funds, or tax incentives in high-risk fields like neuromorphic photonic computing and AI.
- *Mechanisms to Encourage Technology Transfer:* Universities are often the home to emerging technologies, like NPC, and often those technology discoveries end there. The US needs to develop a suite of mechanisms that facilitate technology transfer and licensing from universities to startups and industries willing to take the risk for maturing technology development and bringing new technologies to the market.

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